

## SLAVONSKI BROD, Croatia

WMO: 143700

Lat: 45.1592N

Lon: 17.9953E

Elev: 293

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.4       | 17.7 | 6.8                           | 7.9 | 15.3 | 12.0 | 10.2 | 20.4 | 15.9                  | 43.7 | 13.5 | 39.2 | 2.5                   | 60   | 0.433 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.8                   | 93.4            | 71.7 | 90.4 | 70.8 | 87.4 | 69.9 | 73.9                | 87.9 | 72.5 | 86.5 | 71.1 | 84.6 | 4.5                  | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 69.6                            | 110.2 | 80.0 | 68.0 | 104.0 | 78.7 | 66.5 | 98.6 | 76.9 | 37.6          | 88.3 | 36.3 | 86.7 | 35.1 | 84.9 | 79.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 13.6              | 11.1 | 9.3 | 2.9                        | 98.0 | 9.5                | 3.7 | -3.9                                               | 100.7 | -9.5       | 102.9 | -14.9      | 105.0 | -21.8      | 107.7 |
|                   |      |     | 2.5                        | 76.8 | 9.4                | 1.6 | -4.3                                               | 78.0  | -9.8       | 78.9  | -15.1      | 79.8  | -21.9      | 81.0  |
|                   |      |     | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |     | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 53.6  | 33.4  | 37.1  | 45.5  | 54.4  | 62.0  | 68.5  | 71.9  | 71.7  | 62.6  | 54.2  | 45.0  | 35.5 | (6)  |
| (7)  |                                                                                  | DBStd       | 15.25 | 8.69  | 8.44  | 7.99  | 6.56  | 6.19  | 6.29  | 5.46  | 5.31  | 6.47  | 7.17  | 7.75  | 7.77 | (7)  |
| (8)  |                                                                                  | HDD50       | 1775  | 515   | 363   | 180   | 33    | 3     | 0     | 0     | 0     | 1     | 42    | 185   | 453  | (8)  |
| (9)  |                                                                                  | HDD65       | 4841  | 979   | 781   | 604   | 321   | 131   | 36    | 8     | 9     | 119   | 337   | 601   | 915  | (9)  |
| (10) |                                                                                  | CDD50       | 3079  | 1     | 2     | 40    | 165   | 375   | 554   | 680   | 672   | 380   | 173   | 34    | 3    | (10) |
| (11) |                                                                                  | CDD65       | 671   | 0     | 0     | 0     | 3     | 38    | 140   | 224   | 215   | 48    | 3     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 8182  | 0     | 0     | 4     | 104   | 561   | 1701  | 2620  | 2518  | 595   | 78    | 1     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 3250  | 0     | 0     | 0     | 14    | 127   | 640   | 1159  | 1131  | 175   | 4     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.7   | 3.6   | 4.0   | 4.7   | 4.2   | 4.0   | 3.8   | 3.7   | 3.3   | 3.3   | 3.1   | 3.2   | 3.4  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 32.0  | 2.0   | 1.8   | 2.0   | 2.6   | 3.1   | 3.3   | 3.2   | 2.6   | 3.4   | 2.7   | 2.6   | 2.5  | (15) |
| (16) |                                                                                  | PrecMax     | 40.4  | 3.6   | 3.5   | 3.9   | 6.7   | 8.6   | 7.6   | 7.9   | 8.6   | 8.9   | 6.8   | 4.8   | 5.5  | (16) |
| (17) |                                                                                  | PrecMin     | 17.8  | 0.5   | 0.3   | 0.2   | 0.4   | 0.6   | 1.1   | 0.6   | 0.4   | 0.5   | 0.2   | 0.2   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 6.1   | 0.9   | 0.9   | 0.9   | 1.5   | 1.8   | 1.4   | 1.9   | 2.0   | 1.9   | 1.7   | 1.2   | 1.4  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 59.9  | 64.3  | 73.7  | 81.7  | 87.6  | 93.4  | 97.5  | 98.9  | 90.4  | 80.2  | 71.4  | 61.6 | (19) |
| (20) |                                                                                  |             | MCWB  | 50.2  | 50.8  | 55.4  | 62.0  | 67.2  | 72.9  | 72.4  | 71.5  | 68.3  | 64.2  | 58.7  | 52.3 | (20) |
| (21) |                                                                                  | 2%          | DB    | 54.4  | 59.0  | 69.2  | 76.7  | 83.6  | 90.1  | 93.5  | 93.7  | 84.5  | 76.0  | 65.9  | 54.8 | (21) |
| (22) |                                                                                  |             | MCWB  | 47.3  | 48.0  | 53.5  | 59.3  | 65.9  | 71.6  | 72.1  | 71.0  | 66.6  | 62.4  | 55.4  | 48.5 | (22) |
| (23) |                                                                                  | 5%          | DB    | 49.0  | 54.1  | 65.1  | 72.8  | 80.1  | 87.1  | 90.3  | 90.0  | 80.4  | 72.3  | 61.4  | 50.1 | (23) |
| (24) |                                                                                  |             | MCWB  | 44.4  | 45.9  | 51.7  | 57.3  | 64.2  | 70.5  | 71.4  | 70.1  | 65.0  | 60.7  | 53.8  | 45.9 | (24) |
| (25) |                                                                                  | 10%         | DB    | 44.8  | 49.6  | 60.6  | 68.9  | 76.7  | 83.8  | 86.7  | 86.6  | 76.7  | 67.8  | 57.4  | 46.1 | (25) |
| (26) |                                                                                  |             | MCWB  | 41.7  | 44.1  | 49.8  | 55.3  | 62.6  | 68.8  | 70.0  | 69.3  | 63.6  | 58.7  | 52.0  | 42.9 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 51.0  | 51.9  | 56.5  | 63.4  | 69.5  | 75.6  | 75.7  | 75.6  | 70.5  | 65.7  | 59.6  | 52.7 | (27) |
| (28) |                                                                                  |             | MCDWB | 58.7  | 62.1  | 70.4  | 79.2  | 82.7  | 88.6  | 89.6  | 88.7  | 85.7  | 76.9  | 68.5  | 59.7 | (28) |
| (29) |                                                                                  | 2%          | WB    | 47.9  | 49.3  | 54.3  | 60.7  | 67.2  | 73.3  | 73.9  | 73.4  | 68.2  | 63.4  | 57.1  | 49.2 | (29) |
| (30) |                                                                                  |             | MCDWB | 53.2  | 57.2  | 66.2  | 73.6  | 80.1  | 86.4  | 88.3  | 87.5  | 80.5  | 73.8  | 63.1  | 54.3 | (30) |
| (31) |                                                                                  | 5%          | WB    | 44.7  | 46.9  | 52.4  | 58.4  | 65.4  | 71.6  | 72.6  | 71.9  | 66.5  | 61.4  | 54.7  | 46.0 | (31) |
| (32) |                                                                                  |             | MCDWB | 48.1  | 52.7  | 62.8  | 69.5  | 77.8  | 84.7  | 86.9  | 86.1  | 77.1  | 70.2  | 60.9  | 49.5 | (32) |
| (33) |                                                                                  | 10%         | WB    | 42.2  | 44.6  | 50.5  | 56.6  | 63.7  | 69.9  | 71.2  | 70.4  | 64.8  | 59.5  | 52.0  | 43.3 | (33) |
| (34) |                                                                                  |             | MCDWB | 44.9  | 49.0  | 59.4  | 66.4  | 74.6  | 81.8  | 84.5  | 83.9  | 74.1  | 66.5  | 56.9  | 45.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 13.5  | 16.2  | 20.5  | 22.4  | 22.7  | 22.9  | 23.8  | 25.2  | 22.2  | 21.2  | 16.3  | 12.8 | (35) |
| (36) |                                                                                  |             | MCDBR | 22.4  | 26.8  | 30.7  | 31.7  | 29.7  | 28.6  | 30.7  | 32.6  | 30.1  | 28.6  | 25.5  | 21.7 | (36) |
| (37) |                                                                                  |             | MCWBR | 15.2  | 16.8  | 16.4  | 15.4  | 13.6  | 12.2  | 11.7  | 12.6  | 13.7  | 15.9  | 15.8  | 15.6 | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 20.2  | 23.7  | 27.1  | 26.8  | 26.4  | 26.5  | 27.6  | 28.2  | 26.2  | 26.3  | 22.8  | 20.0 | (38) |
| (39) |                                                                                  |             | MCWBR | 14.2  | 15.2  | 14.8  | 13.2  | 12.6  | 11.9  | 11.6  | 11.9  | 12.4  | 14.9  | 14.6  | 14.7 | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.340 | 0.368 | 0.401 | 0.457 | 0.444 | 0.442 | 0.444 | 0.446 | 0.430 | 0.397 | 0.360 | 0.340 | (40) |      |
| (41) |                                                                                  | taud        | 2.361 | 2.279 | 2.224 | 2.080 | 2.173 | 2.223 | 2.215 | 2.215 | 2.227 | 2.315 | 2.370 | 2.370 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 240   | 253   | 261   | 255   | 263   | 264   | 261   | 256   | 248   | 240   | 229   | 226   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 25    | 32    | 39    | 49    | 46    | 44    | 43    | 42    | 38    | 31    | 24    | 22    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 398   | 630   | 1017  | 1400  | 1725  | 1921  | 1949  | 1737  | 1195  | 816   | 474   | 324   | (44) |      |
| (45) |                                                                                  | RadStd      | 61    | 92    | 131   | 152   | 150   | 160   | 141   | 184   | 144   | 92    | 51    | 53    | (45) |      |

## Historical Trends

|      |                         | DBAvg |     | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|-----|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A | N/A     | N/A    | +0.85   | +1.30 | N/A   | N/A         | N/A   | N/A   | (46)  |
| (47) | Regional (35 neighbors) | +1.15 | N/A | N/A     | N/A    | +0.72   | N/A   | -194  | -283        | +180  | +108  | (47)  |

Nomenclature: See separate page